

POWER BI ASSIGNMENT 1 :

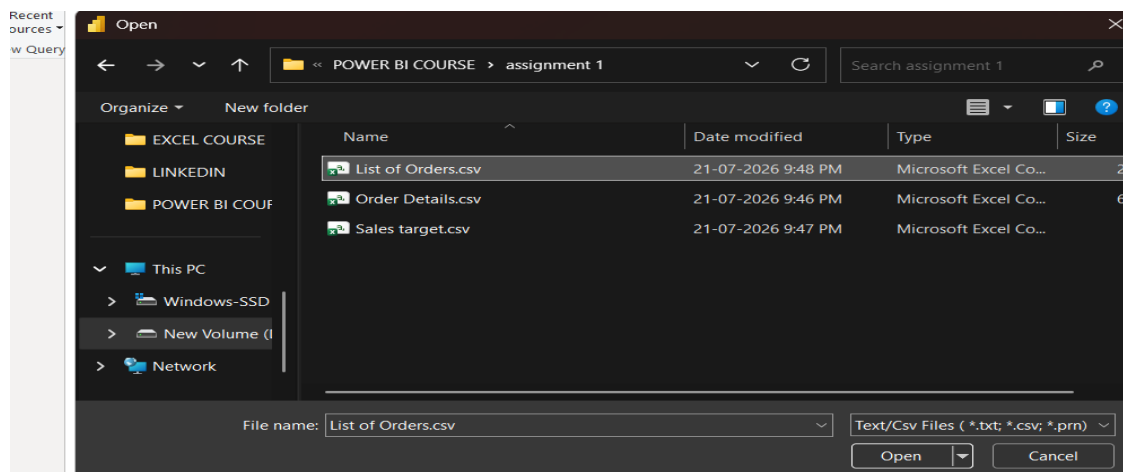
DATA TRANSFORMATION & DATA MODELING

DATE OF SUBMISSION : 24/07/2026 – 08:00AM

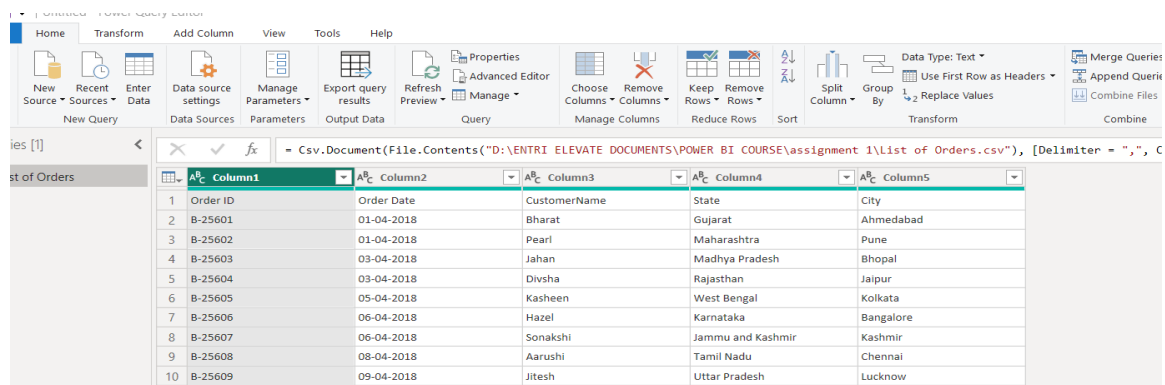
PREVIEW SCREENSHOTS : QUESTIONS

1) IMPORT DATA:

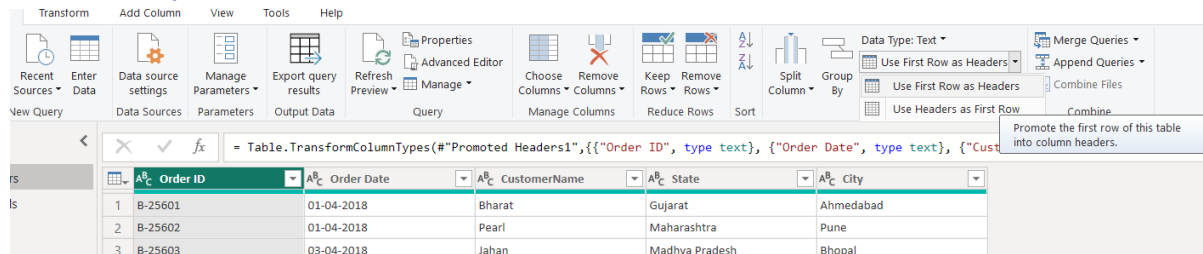
1.1) IMPORT “LIST OF ORDERS.CSV” INTO POWER BI



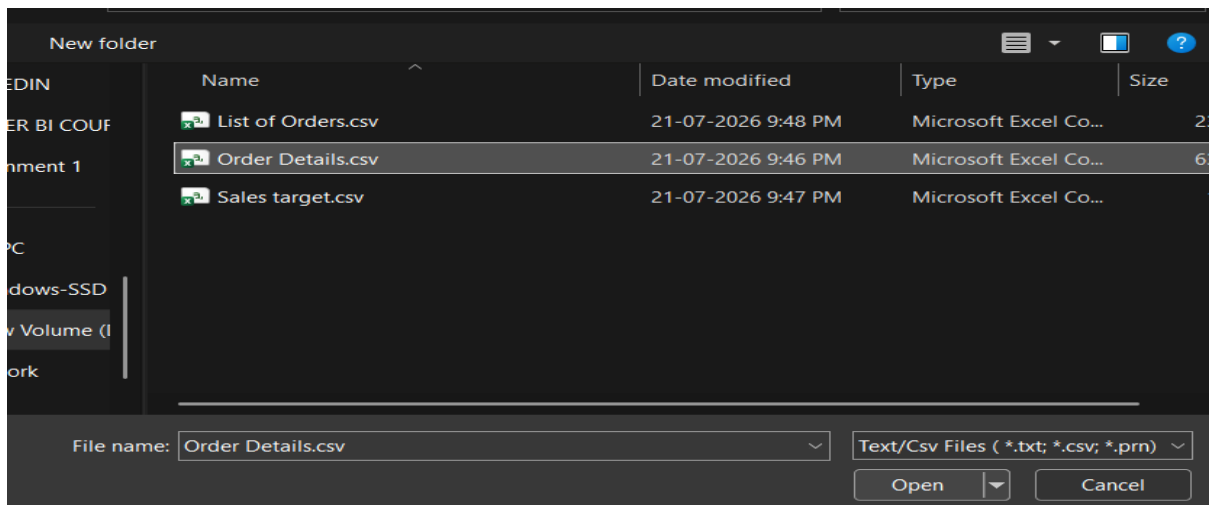
1.1.1) OPEN “LIST OF ORDERS” IN POWER QUERY EDITOR BY CLICKING ON “TRANSFORM”.



1.1.2) “USING FIRST ROW AS HEADER” - IN “LIST OF ORDERS”



1.2) IMPORT “ORDER DETAILS.CSV” INTO POWER QUERY EDITOR.

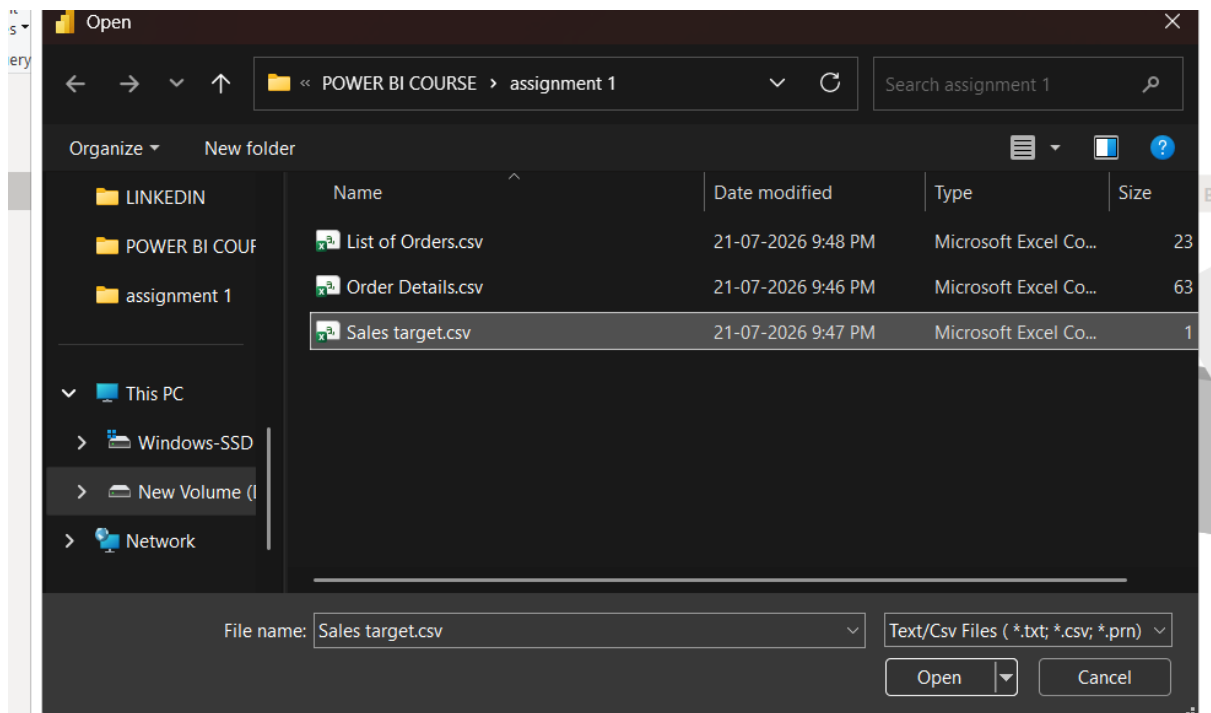


1.2.1) OPEN “ORDER DETAILS.CSV” IN POWER QUERY EDITOR BY CLICKING ON ‘TRANSFORM’.

Table.TransformColumnTypes(#"Promoted headers", {"Order ID", type text}, {"Amount", Int64.Type}, {"Profit", Int64.Type}, {"Quantity", Int64.Type}, {"Category", type text}, {"Sub-Category", type text})

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1275	-1148	7	Furniture	Bookcases
B-25601	66	-12	5	Clothing	Stole
B-25601	8	-2	3	Clothing	Hankerchief
B-25601	80	-56	4	Electronics	Electronic Games
B-25602	168	-111	2	Electronics	Phones
B-25602	424	-272	5	Electronics	Phones
B-25602	2617	1151	4	Electronics	Phones
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	1355	-60	5	Clothing	Trousers
B-25603	24	-30	1	Furniture	Chairs
B-25603	193	-166	3	Clothing	Saree
B-25603	180	5	3	Clothing	Trousers
B-25603	116	16	4	Clothing	Stole
B-25603	107	36	6	Clothing	Stole
B-25603	12	1	2	Clothing	Hankerchief
B-25603	38	18	1	Clothing	Kurti
B-25604	65	17	2	Clothing	T-shirt
B-25604	157	5	9	Clothing	Saree
B-25605	75	0	7	Clothing	Saree
B-25606	87	4	2	Clothing	Shirt
B-25607	50	15	4	Clothing	Leggings
B-25608	1364	-1864	5	Furniture	Tables
B-25608	476	0	3	Furniture	Chairs
B-25608	257	23	5	Clothing	Hankerchief
B-25608	856	385	6	Electronics	Printers
B-25609	485	24	4	Electronics	Electronic Games

1.3) IMPORT "SALES TARGET.CSV" INTO POWER QUERY EDITOR.



1.3.1) OPEN "SALES TARGET.CSV" IN POWER QUERY EDITOR BY CLICKING ON 'TRANSFORM'.

The screenshot shows the Power Query Editor interface. The 'Transform' tab is active. The data table is displayed with the following columns: 'Month of Order Date', 'Category', and 'Target'. The formula bar shows the query name 'Table.TransformColumnTypes' and the formula: `Table.TransformColumnTypes(#"Promoted headers", {{"Month of Order Date", type date}, {"Category", type text}, {"Target", Int64.Type}})`.

	Month of Order Date	Category	Target
1	18-04-2026	Furniture	10400
2	18-05-2026	Furniture	10500
3	18-06-2026	Furniture	10600
4	18-07-2026	Furniture	10800
5	18-08-2026	Furniture	10900
6	18-09-2026	Furniture	11000
7	18-10-2026	Furniture	11100
8	18-11-2026	Furniture	11300
9	18-12-2026	Furniture	11400
10	19-01-2026	Furniture	11500
11	19-02-2026	Furniture	11600
12	19-03-2026	Furniture	11800
13	18-04-2026	Clothing	12000
14	18-05-2026	Clothing	12000
15	18-06-2026	Clothing	12000
16	18-07-2026	Clothing	14000
17	18-08-2026	Clothing	14000
18	18-09-2026	Clothing	14000
19	18-10-2026	Clothing	16000
20	18-11-2026	Clothing	16000
21	18-12-2026	Clothing	16000
22	19-01-2026	Clothing	16000
23	19-02-2026	Clothing	16000
24	19-03-2026	Clothing	16000
25	18-04-2026	Electronics	9000
26	18-05-2026	Electronics	9000

2) DATA TRANSFORMATION:

2.1) DATA TRANSFORM IN "LIST OF ORDERS".

2.1.1) "LIST OF ORDERS" TABLE TO ONLY THE FIRST 500 ROWS.

Queries [3] X ✓ fx = Table.TransformColumnTypes("#Promoted Headers1",{{"Order ID", type text}, {"Order Date", type text}, {"CustomerName", type text},

Order ID	Order Date	CustomerName	State	City
B-25869	19-11-2018	Parakh	Nagaland	Kohima
B-25870	20-11-2018	Pranav	Andhra Pradesh	Hyderabad
B-25871	21-11-2018	Gunjal	Gujarat	Surat
B-25872	22-11-2018	Saurabh	Maharashtra	Mumbai
B-25873	23-11-2018	Divyansha	Madhya Pradesh	Indore
B-25874	24-11-2			
B-25875	24-11-2			
B-25876	24-11-2			
B-25877	24-11-2			
B-25878	24-11-2			
B-25879	24-11-2			
B-25880	24-11-2			
B-25881	25-11-2			
B-25882	26-11-2			
B-25883	27-11-2			
B-25884	28-11-2018	Sumeet	Maharashtra	Mumbai
B-25885	28-11-2018	Shatayu	Madhya Pradesh	Indore
B-25886	28-11-2018	Brijesh	Rajasthan	Udaipur
B-25887	01-12-2018	Vedant	Uttar Pradesh	Allahabad
B-25888	02-12-2018	Bahar	Bihar	Amritsar

Keep Top Rows

Specify how many rows to keep.

Number of rows

500

OK Cancel

2.1.2) "ORDER DATE" COLUMN IN THE "LIST OF ORDERS" TABLE IS SET TO DATA TYPE 'DATE'.

Table.TransformColumnTypes("#Changed Type with Locale",{{"Order Date", type date}})

Order Date	CustomerName	State	City
01-04-2018	Bharat	Gujarat	Ahmedabad
01-04-2018	Pearl	Maharashtra	Pune
03-04-2018	Jahan	Madhya Pradesh	Bhopal
05-04-2018	Divyansha	Rajasthan	Jaipur
22-04-2018	Atharv	West Bengal	Kolkata
22-04-2018	Vini	Karnataka	Bangalore

Change Type with Locale

Change the data type and select the locale of origin.

Data Type

Date

Locale

English (United Kingdom)

Sample input values:

29/03/2016

29 March 2016

29 March

March 2016

OK Cancel

2.1.3) VIEW THE CHANGED "DATE" OF "ORDER DATE"

IGNITION 1 TRANSFORM

Add Column View Tools Help

Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Reduce Rows Sort Split Column Group By Data Type: Date Use First Row as Headers Replace Values Transform

fx = Table.TransformColumnTypes("#Changed Type1", {{"Order Date", type date}}, "en-GB")

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divyansha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata

2.1.4) "AMOUNT" AND "TARGET" COLUMNS TO 'FIXED DECIMAL NUMBER'.

POWER BI ASSIGNMENT 1 TRANSFORM

Transform Add Column View Tools Help

Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By

Data Type: Fixed decimal number

Decimal Number

Fixed decimal number

Whole Number

Percentage

Date/Time

Date

Time

Date/Time/Timezone

Duration

Text

Currency.Type

Category

is

Chief

c Games

Order ID	Amount	Profit	Quantity	Category
1 B-25601	1,275.00	-1,148.00	7	Furniture
2 B-25601	66.00	-12.00	5	Clothing
3 B-25601	8.00	-2.00	3	Clothing
4 B-25601	80.00	-56.00	4	Electronics

Transform Add Column View Tools Help

Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By

Data Type: Fixed decimal number

Decimal Number

Fixed decimal number

Whole Number

Percentage

Date/Time

Date

Time

Date/Time/Timezone

Duration

Text

Currency.Type

Category

is

Chief

c Games

True/False

Order ID	Amount	Profit	Quantity	Category
1 B-25601	1,275.00	-1,148.00	7	Furniture
2 B-25601	66.00	-12.00	5	Clothing
3 B-25601	8.00	-2.00	3	Clothing
4 B-25601	80.00	-56.00	4	Electronics
5 B-25602	168.00	-111.00	2	Electronics

2.1.5) "CUSTOMERNAME" COLUMN INTO PROPER CASE, ENSURING CONSISTENT "CAPITALIZATION FOR EACH WORD".

POWER BI ASSIGNMENT 1 TRANSFORM

Transform Add Column View Tools Help

Transpose Inverse Rows Pivot Rows Data Type: Text Replace Values Unpivot Columns Detect Data Type Fill Move Merge Columns Rename Pivot Column Convert to List Split Column Format

Format

lowercase

UPPERCASE

Capitalize Each Word

Trim

Clean

Add Prefix

Add Suffix

Statistics Standard Deviation

ABC 123 Extract Parse

Table.TransformColumns(#"Changed Type with Locale1",{{"CustomerName", Text.Clean}}

Order ID	Order Date	CustomerName	State
1 B-25601	01-04-2018	Bharat	
2 B-25602	01-04-2018	Pearl	
3 B-25603	03-04-2018	Jahan	
4 B-25604	03-04-2018	Divsha	Rajasthan
5 B-25605	05-04-2018	Karbhaj	West Bengal

VIEW THE PROPER CASE OF "CUSTOMER NAME"

Table.TransformColumns(#"Changed Type with Locale1",{{"CustomerName", Text.Clean}}

Order Date	CustomerName	State
03-08-2018	Daksh	Haryana
03-08-2018	Rane	Maharash
03-08-2018	Navdeep	Madhya P
03-08-2018	Ashwin	Goa
07-08-2018	Aman	Nagaland
08-08-2018	Devendra	Andhra Pr
09-08-2018	Kartik	Gujarat
10-08-2018	Shivam	Maharash
11-08-2018	Harsh	Madhya P
12-08-2018	Nitant	Rajasthan
13-08-2018	Ayush	Maharash
14-08-2018	Priyanshu	Madhya P
14-08-2018	Nishant	Maharash
14-08-2018	Vaibhav	Madhya P
17-08-2018	Shivam	Uttar Prac

2.1.6) MERGE THE "STATE" AND "CITY" COLUMNS TO CREATE A NEW COLUMN NAMED "LOCATION".

Table.TransformColumns("#Capitalized Each Word",{{"CustomerName", Text.Proper, type text}})

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604				
B-25605				
B-25606				
B-25607				
B-25608				
B-25609				
B-25610				
B-25611				
B-25612				
B-25613				
B-25614				
B-25615				
B-25616				
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune

Merge Columns

Choose how to merge the selected columns.

Separator:

New column name (optional):

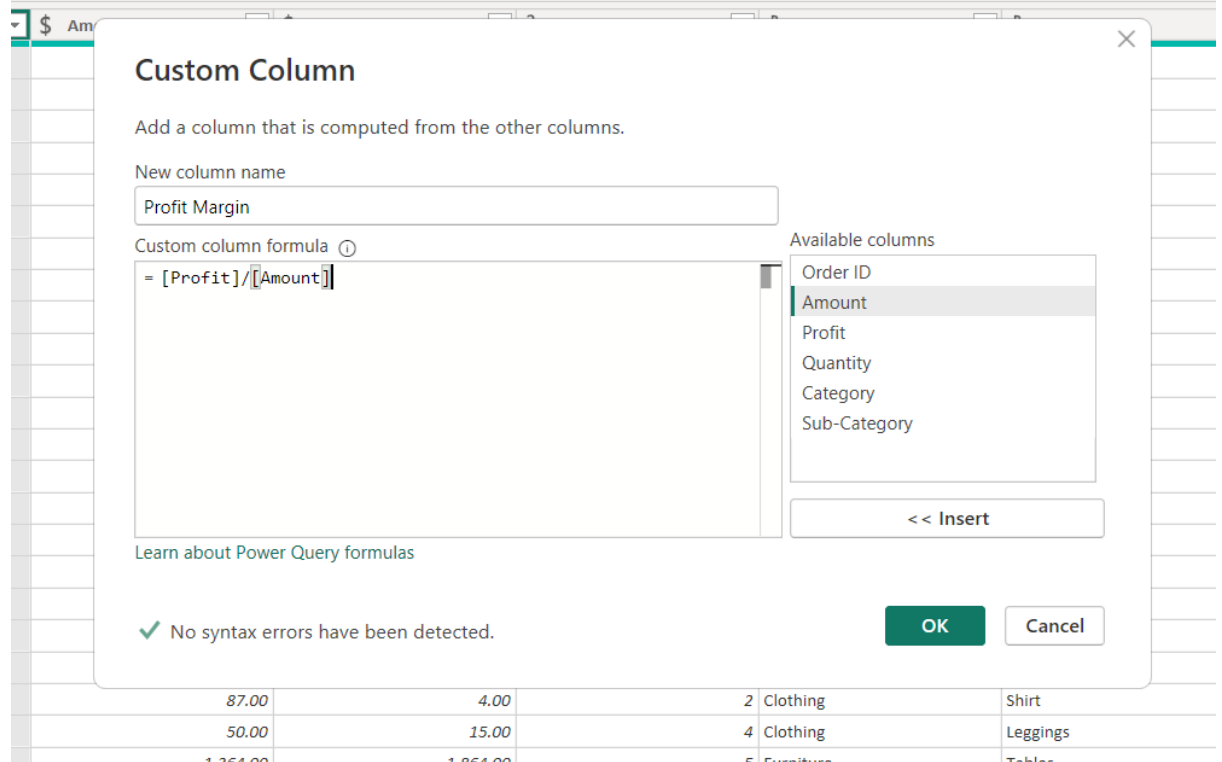
VIEW OF MERGE COLUME AS "LOCATION".

e", "Location"}})

Location
Ahmedabad,Gujarat
Pune,Maharashtra
Bhopal,Madhya Pradesh
Jaipur,Rajasthan
Kolkata,West Bengal
Bangalore,Karnataka
Kashmir,Jammu and Kashmir
Chennai,Tamil Nadu
Lucknow,Uttar Pradesh
Patna,Bihar
Thiruvananthapuram,Kerala
Chandigarh,Punjab
Chandigarh,Haryana
Simla,Himachal Pradesh
Gangtok,Sikkim
Goa,Goa
Kohima,Nagaland

2.1.7) CREATE A NEW "CUSTOM COLUMN" NAMED "PROFIT MARGIN" AS THE PERCENTAGE OF "PROFIT" DIVIDED BY "AMOUNT".

transformColumnTypes({# Promoted headers ,{{ Order ID , type text}}, { Amount , Currency.type}, { Profit , Currency.type},



Custom Column

Add a column that is computed from the other columns.

New column name

Profit Margin

Custom column formula ⓘ

= [Profit]/[Amount]

Available columns

- Order ID
- Amount
- Profit
- Quantity
- Category
- Sub-Category

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

87.00	4.00	2	Clothing	Shirt
50.00	15.00	4	Clothing	Leggings
1 354.00	1 854.00	5	Furniture	Tables

CHANGE TYPOS IN "PROFIT MARGIN" COLUMNS TO 'FIXED DECIMAL NUMBER'.

Sub-Category	\$ Profit Margin
cases	1.2
	\$
erchief	123
onic Games	%
es	Date/Time
es	Date
es	Time
	Date/Time/Timezone
	Duration
ers	Text
s	True/False
	Binary
ers	Using Locale...
	0.34

\$ Profit Margin
-0.90
-0.18
-0.25
-0.70
-0.66
-0.64
0.44
0.38
-0.04
-0.04
-1.25
-0.86
0.03

2.1.8) NEW CONDITIONAL COLUMN NAMED "PROFIT STATUS".

if [Profit] < 0 then "Loss"

else if [Profit] = 0 then "Break-Even"

else if [Profit] > 0 then "Profit"

Table.TransformColumnTypes("#Added Custom",{{"Profit Margin", Currency.Type}})

\$ Amount	\$ Profit	123 Quantity	ABC Category	ABC Sub-Category	\$ Profit Margin
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Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name: Profit Status

	Column Name	Operator	Value		Output
If	Profit	is less than	0	Then	Loss
Else If	Profit	equals	0	Then	Break-Even
Else If	Profit	is greater than	0	Then	Profit

Add Clause

Else

OK Cancel

CHANGE TYPOS IN "PROFIT STATUS" COLUMNS TO 'TEXT'.

profit] = 0 then "Break-Even" else if

Profit Margin

-0.90

-0.18

-0.25

-0.70

-0.66

-0.64

0.44

0.38

-0.04

-0.04

-1.25

-0.86

0.03

0.14

ABC 123

Decimal Number

Fixed decimal number

Whole Number

Percentage

Date/Time

Date

Time

Date/Time/Timezone

Duration

Text

True/False

Binary

Using Locale...

ABC

Profit Status

-0.90

-0.18

-0.25

-0.70

-0.66

-0.64

0.44

0.38

-0.04

-0.04

-1.25

-0.86

0.03

0.14

0.34

0.08

0.47

Loss

Loss

Loss

Loss

Loss

Loss

Profit

Profit

Loss

Loss

Loss

Loss

Profit

Profit

Profit

Profit

Profit

3) MERGING DATA (JOINS):

3.1) MERGE THE "LIST OF ORDERS" AND "ORDER DETAILS" TABLES INTO A NEW SINGLE TABLE NAMED "ORDERS DATA" BASED ON THE "ORDER ID" RELATIONSHIP "LEFT OUTER".

Select a table and matching columns to create a merged table.

List of Orders

Order ID	Order Date	CustomerName	City, State
B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat
B-25602	01-04-2018	Pearl	Pune,Maharashtra
B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh
B-25604	03-04-2018	Divsha	Jaipur,Rajasthan
B-25605	05-04-2018	Kasheen	Kolkata,West Bengal

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1,148.00	7	Furniture	Bookcases	-0.90	Loss
B-25601	66.00	-12.00	5	Clothing	Stole	-0.18	Loss
B-25601	8.00	-2.00	3	Clothing	Hankerchief	-0.25	Loss
B-25601	80.00	-56.00	4	Electronics	Electronic Games	-0.70	Loss
B-25602	168.00	-111.00	2	Electronics	Phones	-0.66	Loss

Join Kind

Left Outer (all from first, matching from second)

☒ Use fuzzy matching to perform the merge

▷ Fuzzy matching options

VIEW THE "ORDERS DATE".

Close & Apply	New Source	Recent Sources	Enter Data	Data source settings	Manage Parameters	Export query results	Refresh Preview	Advanced Editor	Choose Columns	Remove Columns	Keep Rows	Remove Rows	Split Column	Group By	Use First Row as Headers	Append Queries
Close	New Query			Data Sources	Parameters	Output Data	Query	Manage	Manage Columns		Reduce Rows		Sort	Transform	Replace Values	Combine Files
Queries [3]																
= Table.ExpandTableColumn(#"Merged Queries", "Order Details", {"Order ID", "Amount", "Profit", "Quantity", "Category", "Sub-Category",																
Orders Data	Order ID	Order Date	CustomerName	City, State	Order Details.Order ID	Order Details.Amount										
Order Details	1	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25601	1,275.00									
Sales target	2	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25601	66.00									
	3	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25601	8.00									
	4	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25601	80.00									
	5	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	200.00									
	6	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	44.00									
	7	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	7.00									
	8	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	11.00									
	9	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	16.00									
	10	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	172.00									
	11	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	49.00									
	12	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	823.00									
	13	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	23.00									
	14	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-25651	457.00									
	15	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-26051	184.00									
	16	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-26051	676.00									
	17	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-26051	669.00									
	18	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-26051	80.00									
	19	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-26051	216.00									
	20	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-26051	85.00									
	21	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-26051	382.00									
	22	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat	B-26051	490.00									

4) HANDLING MISSING DATA & DUPLICATE DATA:

4.1) CHECK FOR "REMOVE DUPLICATE" IN EACH TABLES

Table	Any Column	Text
	✕ ✓ fx = Table.Distinct(#"Expanded Order Details", {"Or	
	ABC Order ID 1 B-25601 2 B-25602 3 B-25603 4 B-25604 5 B-25605 6 B-25606 7 B-25607 8 B-25608 9 B-25609 10 B-25610 11 B-25611 12 B-25612 13 B-25613 14 B-25614 15 B-25615 16 B-25616 17 B-25617 18 B-25618 19 B-25619 20 B-25620 Copy Remove Remove Other Columns Duplicate Column Add Column From Examples... Remove Duplicates Remove Errors Change Type Transform Replace Values... Replace Errors... Split Column Group By... Fill Unpivot Columns Unpivot Other Columns Unpivot Only Selected Columns Rename... Move Drill Down	CustomerName rat rl an sha heen el akshi ushi sh esh ta chand kesh dana vna ak ar nju nesh ta

NO DUPLICATE VALUES

✕ ✓ fx let columnNames = {"Order ID"}, addCount = Table.Group(#"Removed Duplicates", columnNames, {"Count", Table.RowCount, type number}),	Order ID Order Date CustomerName City, State Order Details.Order ID Order Details.Amount Ord	Query Settings PROPERTIES Name Orders Data All Properties APPLIED STEPS Source Promoted Headers Changed Type Kept First Rows Changed Type with Locale Changed Type1 Changed Type with Locale1 Capitalized Each Word Merged Columns Merged Queries Expanded Order Details Removed Duplicates Kept Duplicates
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5) SORTING AND FILTERING DATA:

5.1) 'ORDERS DATA' TABLE, SORTING AND FILTERING THE "ORDER DATE", "STATE"

AND "CATEGORY".

5.1.1) SORTING "ORDER DATE".

Table.Sort(#"Filtered Rows",{{"Order Date", Order.Descending}})

Order ID	Order Date	CustomerName	City, State	Order Details.Order ID	Order Details.Amount	Order Details.Profit
Hitika	Indore,Madhya Pradesh	B-26100		828.00		
Bhishm	Mumbai,Maharashtra	B-26099		9.00		
Pinky	Kashmir,Jammu and Kashmir	B-26098		82.00		
Vini	Bangalore,Karnataka	B-26097		97.00		
Atharv	Kolkata,West Bengal	B-26096		45.00		
Monisha	Jaipur,Rajasthan	B-26095		6.00		
Deepak	Bhopal,Madhya Pradesh	B-26094		152.00		
Manju	Hyderabad,Andhra Pradesh	B-26091		158.00		
Ramesh	Ahmedabad,Gujarat	B-26092		97.00		
Sarita	Pune,Maharashtra	B-26093		33.00		
Sagar	Kohima,Nagaland	B-26090		80.00		
Bhavana	Gangtok,Sikkim	B-26088		11.00		
Mukesh	Chandigarh,Haryana	B-26086		43.00		
Vandana	Simla,Himachal Pradesh	B-26087		119.00		
Shrichand	Chandigarh,Punjab	B-26085		86.00		
Kanak	Goa,Goa	B-26089		59.00		
Anita	Thiruvananthapuram,Kerala	B-26084		209.00		
Yogesh	Patna,Bihar	B-26083		43.00		
Iltesh	Lucknow,Uttar Pradesh	B-26082		95.00		
Kasheen	Kolkata,West Bengal	B-26078		17.00		
Divsha	Jaipur,Rajasthan	B-26077		62.00		
Sonakshi	Kashmir,Jammu and Kashmir	B-26080		109.00		
Hazel	Bangalore,Karnataka	B-26079		18.00		
Aarushi	Chennai,Tamil Nadu	B-26081		359.00		
Bharat	Ahmedabad,Gujarat	B-26074		57.00		
Pournamasi	Indore,Madhya Pradesh	B-26073		37.00		

5.1.2) FILTERING "LOCATION" – "TAMIL NADU".

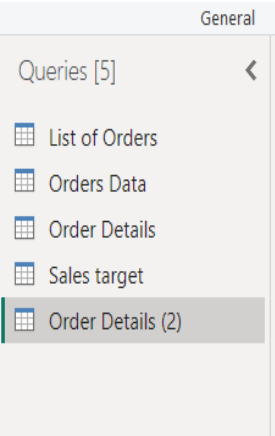
Rows(#"Renamed Columns1", each ([Location] = "Chennai,Tamil Nadu"))

Order Date	CustomerName	Location	Amount	Profit
22-03-2019	Aarushi	Chennai,Tamil Nadu	359.00	
14-02-2019	Aarushi	Chennai,Tamil Nadu	326.00	
09-02-2019	Kalyani	Chennai,Tamil Nadu	22.00	
15-11-2018	Akshay	Chennai,Tamil Nadu	112.00	
21-09-2018	Dinesh	Chennai,Tamil Nadu	12.00	
11-07-2018	Surabhi	Chennai,Tamil Nadu	58.00	
23-06-2018	Amisha	Chennai,Tamil Nadu	87.00	
08-04-2018	Aarushi	Chennai,Tamil Nadu	1,364.00	

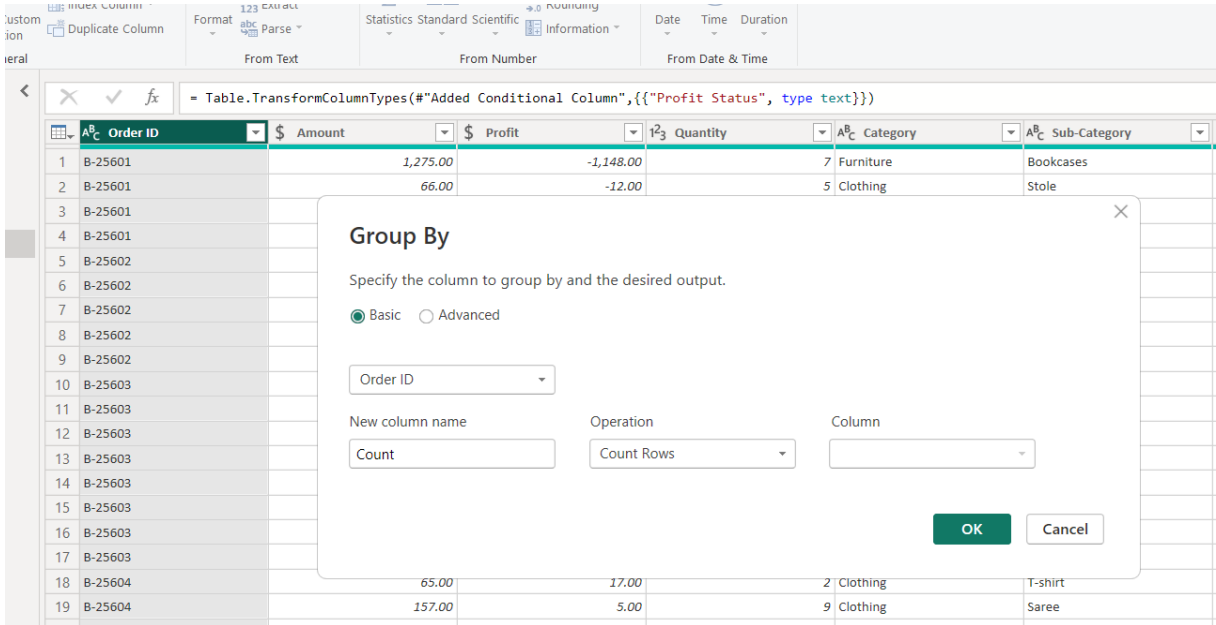
6) GROUPING AND AGGREGATING DATA:

6.1) MAKE DUPLICATE OF "ORDER DETAILS".

6.1.1) DUPLICATE OF “ORDER DETIALS(2)”.



6.1.2) COUNT EACH “ORDER ID”.



VIEW THE COUNT OF “ORDER ID” →

Order ID	Count
B-25601	4
B-25602	5
B-25603	8
B-25604	2
B-25605	1
B-25606	1
B-25607	1
B-25608	4
B-25609	2
B-25610	6
B-25611	1
B-25612	1
B-25613	1
B-25614	2
B-25615	1
B-25616	4
B-25617	1
B-25618	2
B-25619	1
B-25620	1
B-25621	3
B-25622	1
B-25623	4
B-25624	1
B-25625	3
B-25626	2
B-25627	1
B-25628	3

6.1.3) “AVERAGE PROFIT” BY CATEGORY.

FROM TEXT FROM NUMBER FROM DATE & TIME

fx = Table.TransformColumnTypes(#"Added Conditional Column",{{"Profit Status", type text}})

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
B-25601	1,275.00	-1,148.00		Furniture	Bookcases	
B-25601	66.00	-12.00		Clothing	Stole	
B-25601						
B-25602						
B-25602						
B-25602						
B-25603						
B-25603						
B-25603						
B-25603						
B-25603						
B-25603						
B-25603						
B-25604	65.00	17.00		Clothing	T-shirt	

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Category

New column name: Average Profit

Operation: Average

Column: Profit

OK Cancel

VIEW OF "AVERAGE PROFIT".

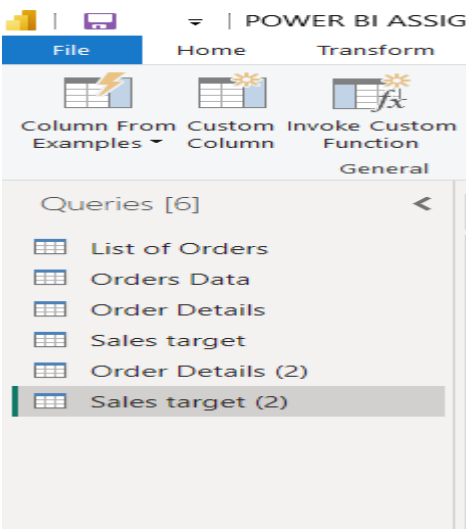
From Text From Number

fx = Table.Group(#"Changed Type1", {"Category

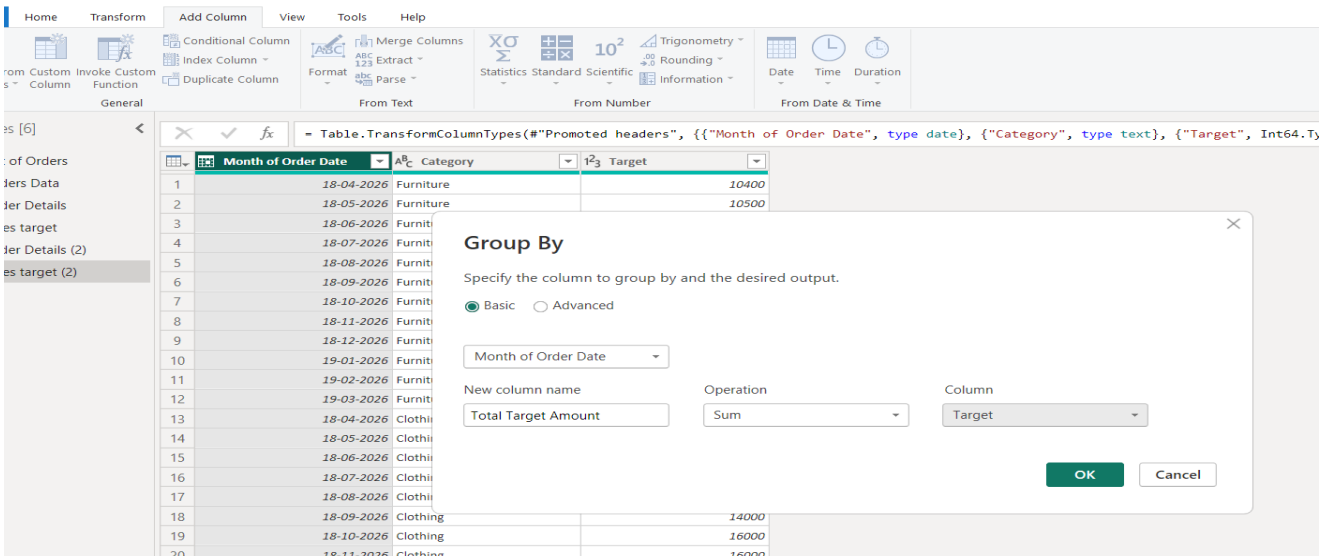
Category	Average Profit
Furniture	9.456790123
Clothing	11.76290832
Electronics	34.07142857

6.2) MAKE DUPLICATE OF "SALES TARGET"

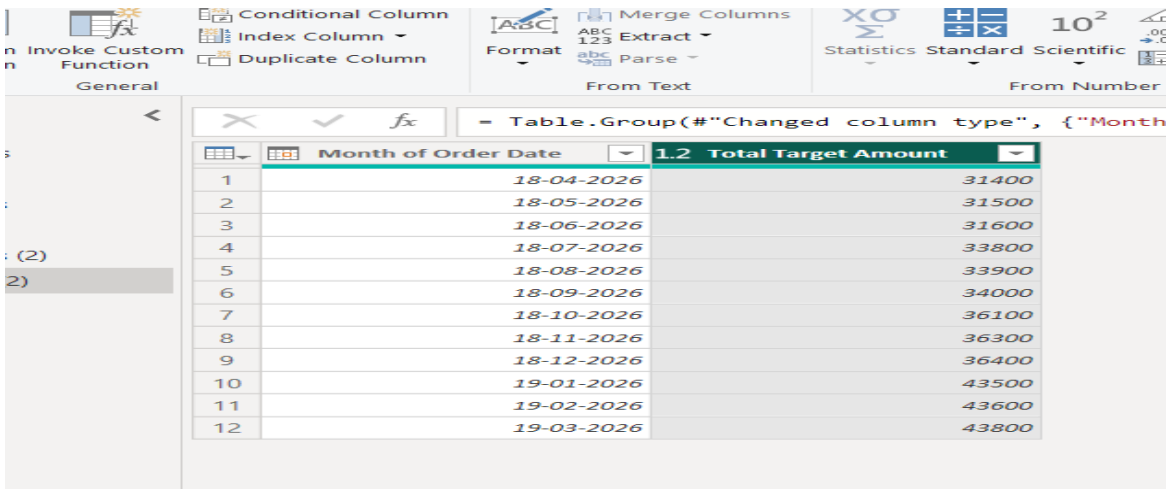
6.2.1) DUPLICATE OF “SALES TARGET(2)”.



6.2.2) “TOTAL TARGET AMOUNT” BY MONTH OF ORDER DATE

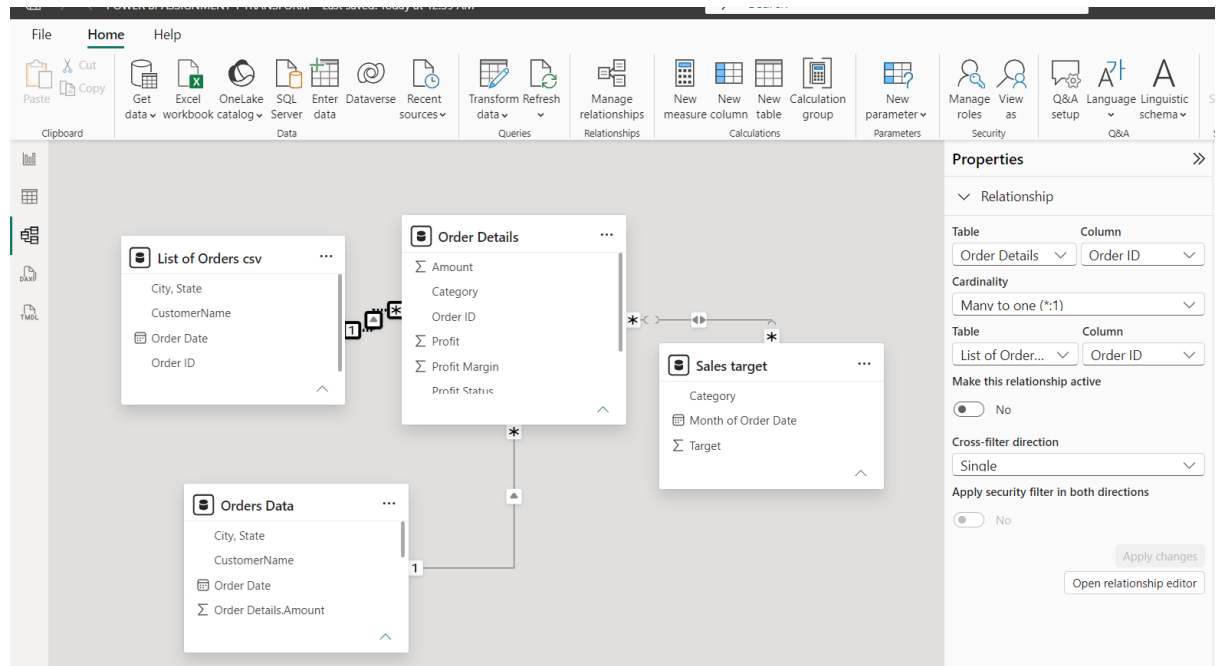


VIEW THE “TOTAL TARGET AMOUNT”

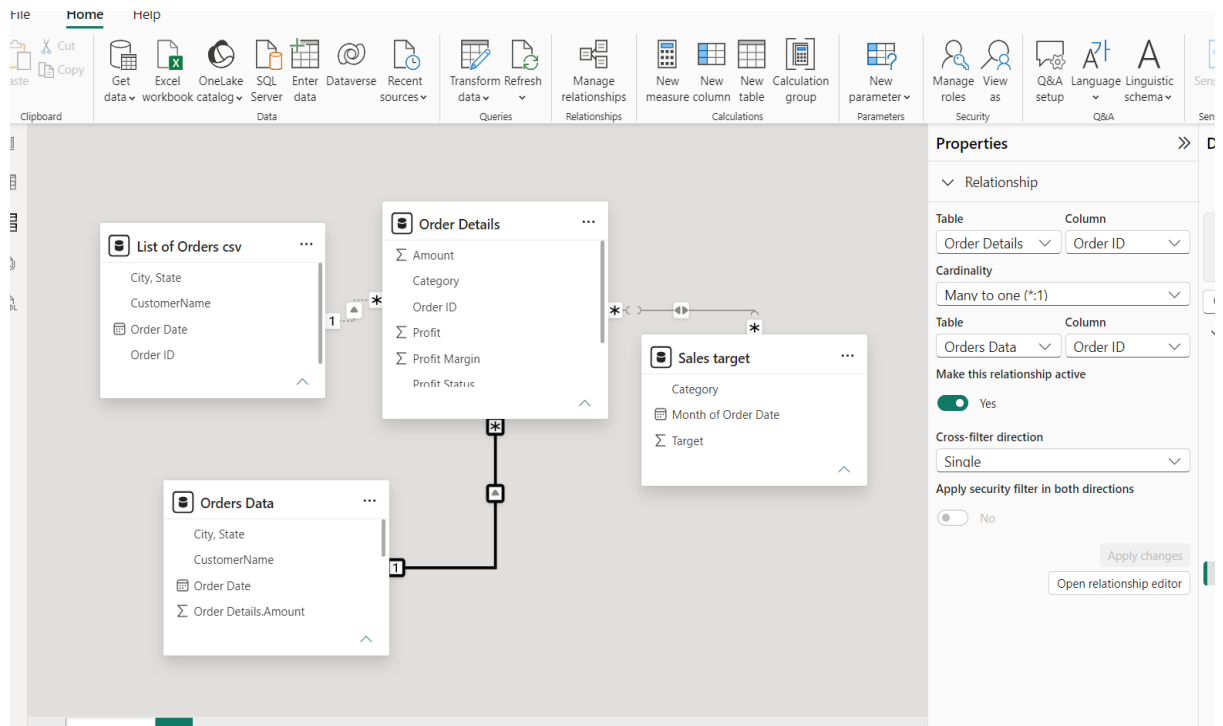


7) DATA MODELING:

7.1) DATA MODELING “LIST OF ORDERS” AND “ORDER DETAILS” TABLES USING THE ‘ORDER ID’.

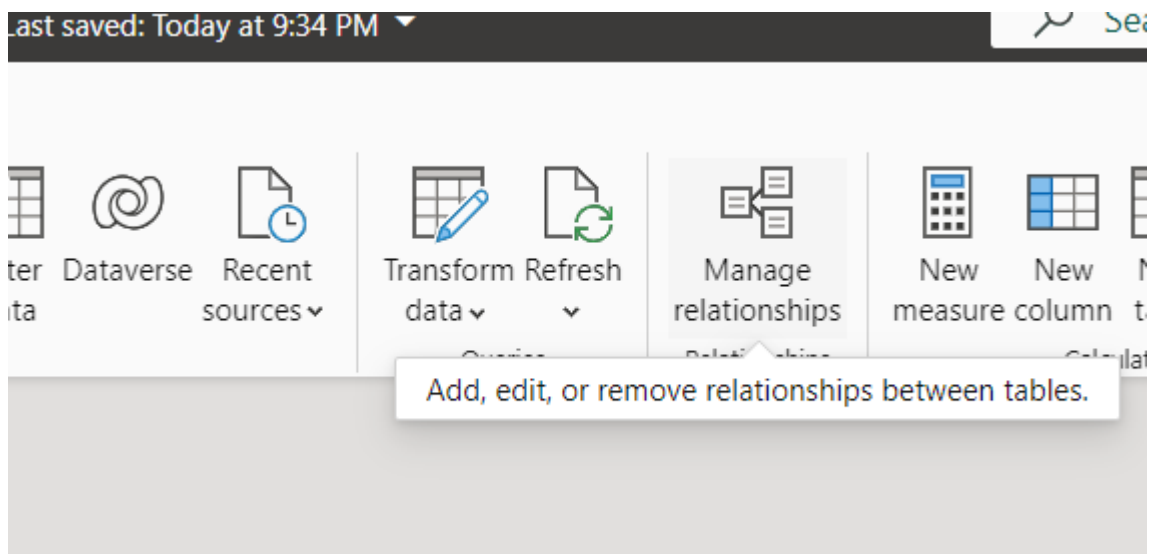


7.2) DATA MODELING “ORDER DETAILS” AND “ORDERS DATE” TABLES USING THE ‘ORDER ID’.



7.2.1) MANAGE RELATIONSHIPS TO “INACTIVE” TO “ACTIVE”.

STEP 1 - CLICK TO "MANAGE RELATIONSHIPS"



STEP 2 – CHECK ANY "INACTIVE" THAT MAKE TO "ACTIVE"

From: table (column) ↑	Relationship	To: table (column)	Status	
Order Details (Category)		Sales target (Category)	Active	...
Order Details (Order ID)		Orders Data (Order ID)	Active	...
Order Details (Order ID)		List of Orders (Order ID)	Inactive	...

Edit
 Delete
Switch status to active

STEP 3 – WE CHANGE TO "ACTIVE"

Manage relationships

+ New relationship

Autodetect

Edit

Delete

Filter

☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ Order Details (Category)		Sales target (Category)	Active	...
☐ Order Details (Order ID)		Orders Data (Order ID)	Active	...
☐ Order Details (Order ID)		List of Orders (Order ID)	Active	...

7.3) DATA MODELING "ORDER DETAILS" AND "SALES TARGET" TABLES USING THE

'ORDER ID'.

The screenshot displays the Microsoft Power BI Desktop interface. The top ribbon includes tabs for 'Data', 'Queries', 'Relationships', 'Calculations', and 'Parameters'. The main workspace shows a data model with four tables:

- List of Orders csv**: Columns include City, State, CustomerName, Order Date, and Order ID.
- Order Details**: Columns include Amount, Category, Order ID, Profit, Profit Margin, and Profit Status.
- Orders Data**: Columns include City, State, CustomerName, Order Date, and Order Details.Amount.
- Sales target**: Columns include Category, Month of Order Date, and Target.

Relationships are defined as follows:

- List of Orders csv** (1) to **Order Details** (*).
- Order Details** (*) to **Orders Data** (1).
- Order Details** (*) to **Sales target** (*).

The 'Sales target' table has a warning icon indicating a Many-Many relationship. The right-hand 'Properties' pane shows the relationship settings for 'Order Details' to 'Sales target':

- Relationship**: Table: Order Details, Column: Category.
- Cardinality**: Many to many (*:*)
- Warning**: This relationship has cardinality Many-Many. This should only be used if it is expected that neither column (Category and Category) contains unique values, and that the significantly different behavior of Many-Many relationships is understood. [Learn more](#)
- Table**: Sales target, **Column**: Category.
- Make this relationship active**: Yes (checked).
- Cross-filter direction**: Both.
- Apply security filter in both directions**: No.

8) VERIFICATION VIA CHARTS

